

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Mern Stack

Roll No :- T010	Name:-Sagar Kandari
Class:-TYIT	Batch:-1
Date of Assignment:-30-06-26	Date/Time of Submission:-30-06-26

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code:-

```
JS prac2A.js > ...
1  function calculateSalary(name, basicSalary) {
2      let bonus = 5000;
3      let totalSalary = basicSalary + bonus;
4
5      console.log("Employee Name:", name);
6      console.log("Basic Salary:", basicSalary);
7      console.log("Bonus:", bonus);
8      console.log("Total Salary:", totalSalary);
9      console.log("Sagar Kandari T010");
10
11 }
12
13 calculateSalary("sagar Knadari T010", 30000);
14
```

Output:-

```
Employee Name: sagar Knadari T010
Basic Salary: 30000
Bonus: 5000
Total Salary: 35000
Sagar Kandari T010
```

Code:

```
JS prac2b.js > ...
1  function generateBill(product, quantity, price) {
2      let totalAmount = quantity * price;
3      let finalBill = totalAmount + (totalAmount * 0.05);
4
5      console.log("Product:", product);
6      console.log("Quantity:", quantity);
7      console.log("Price per item:", price);
8      console.log("Total Bill:", finalBill);
9      console.log("Sagar Kandari S010")
10 }
11
12
13 generateBill("Laptop", 2, 50000);
14
```

Output:

```
Product: Laptop
Quantity: 2
Price per item: 50000
Total Bill: 105000
Sagar Kandari S010
```

b. Work with arrays and objects in JavaScript.

Code:

```
let fruits = ["Mango", "Banana", "Apple", "Cherry"];

let student = {
    name: "Sagar",
    age: 20,
    city: "Mumbai"
};

console.log("Fruits:", fruits);
console.log("Second Fruit:", fruits[1]);
```

```
console.log("Student Name:", student.name);
console.log("Student Age:", student.age);
console.log("Student City:", student.city);
console.log("Sagar Kandari T010");
```

Output:

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac2b.js
Fruits: [ 'Mango', 'Banana', 'Apple', 'Cherry' ]
Second Fruit: Banana
Student Name: Sagar
Student Age: 20
Student City: Mumbai
Sagar Kandari T010
```

c. Demonstrate use of arrow functions.

Code:

```
JS prac2c.js > ...
1  // Addition
2  let add = (a, b) => a + b;
3  console.log("Sum:", add(10, 20));
4
5  // Subtraction
6  let subtract = (a, b) => a - b;
7  console.log("Difference:", subtract(20, 10));
8
9  // Multiplication
10 let multiply = (a, b) => a * b;
11 console.log("Product:", multiply(10, 20));
12 console.log("Sagar Kandari T010");
```

Output:

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac2c.js
Sum: 30
Difference: 10
Product: 200
Sagar Kandari T010
```

d. Write a program using array methods (map, filter, reduce).

Code:

```
JS prac2d.js > ...
2
3
4   let doubled = numbers.map(num => num * 2);
5
6
7   let greater25 = numbers.filter(num => num > 25);
8
9
10  let total = numbers.reduce((sum, num) => sum + num, 0);
11
12  console.log("Original Array:", numbers);
13  console.log("Map (Doubled):", doubled);
14  console.log("Filter (Greater than 25):", greater25);
15  console.log("Reduce (Sum):", total);
16  console.log("Sagar Kandari T010")
```

Output:

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac2d.js
Original Array: [ 10, 20, 30, 48, 50 ]
Map (Doubled): [ 20, 40, 60, 96, 100 ]
Filter (Greater than 25): [ 30, 48, 50 ]
Reduce (Sum): 158
Sagar Kandari T010
```

e. Create a program that manipulates objects and displays output.

Code:

```
// 1. Creating an Object
let student = {
  rollNo: "T010",
  name: "Sagar",
  marks: 85
};

console.log("Original Object:");
console.log(student);

// 2. Accessing Properties
```

```
console.log("\nAccessing Properties:");
console.log("Name:", student.name);
console.log("Marks:", student.marks);

// 3. Updating a Property
student.marks = 92;
console.log("\nAfter Updating Marks:");
console.log(student);

// 4. Adding a New Property
student.grade = "A";
console.log("\nAfter Adding Grade:");
console.log(student);

// 5. Deleting a Property
delete student.rollNo;
console.log("\nAfter Deleting Roll Number:");
console.log(student);

// Final Output
console.log("Sagar Kandari T010");
```

Output:

```
PS C:\Users\Sagar_Kandari\OneDrive\Documents\MERN> node prac2e.js

Accessing Properties:
Name: Sagar
Marks: 85

After Updating Marks:
{ rollNo: 'T010', name: 'Sagar', marks: 92 }

After Adding Grade:
{ rollNo: 'T010', name: 'Sagar', marks: 92, grade: 'A' }

After Deleting Roll Number:
{ name: 'Sagar', marks: 92, grade: 'A' }
Sagar Kandari T010
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code:

```
// Array of Students
let students = [
  {
    rollNo: 101,
    name: "Sagar",
    marks: 85
  },
  {
    rollNo: 102,
    name: "SOM",
    marks: 90
  }
];

// Function to Display Student Details
function display(s) {
  console.log("Student Details");
  console.log("-----");

  for (let stud of s) {
    console.log(`Roll No: ${stud.rollNo}`);
    console.log(`Name    : ${stud.name}`);
    console.log(`Marks   : ${stud.marks}`);
    console.log("-----");
  }

  console.log("Sagar Kandari T010");
}

// Function Call
display(students);
```

Output:

```
PS C:\Users\Sagar Kandari\OneDrive\Documents\MERN> node prac2f.js
Student Details
-----
Roll No: 101
Name   : Sagar
Marks  : 85
-----
Roll No: 102
Name   : SOM
Marks  : 90
-----
Sagar Kandari T010
```